

4. Decision Tree Classification

Software :WEKA

Steps

1. Open WEKA → Explorer.
2. Go to Preprocess.
3. Open the dataset.
4. Click Classify.
5. Click Choose.
6. Select:

Trees → J48

7. Select **10-fold cross-validation** if no separate test data is specified.
8. Click **Start**.
9. Record the accuracy.
10. Record the confusion matrix.
11. View the generated decision tree.

Programming code : weather_nominal.arff

```
@relation weather
@attribute outlook {sunny,overcast,rainy}
@attribute temperature {hot,mild,cool}
@attribute humidity {high,normal}
@attribute windy {TRUE,FALSE}
@attribute play {yes,no}

@data
sunny,hot,high,FALSE,no
sunny,hot,high,TRUE,no
overcast,hot,high,FALSE,yes
rainy,mild,high,FALSE,yes
rainy,cool,normal,FALSE,yes
rainy,cool,normal,TRUE,no
overcast,cool,normal,TRUE,yes
sunny,mild,high,FALSE,no
sunny,cool,normal,FALSE,yes
rainy,mild,normal,FALSE,yes
sunny,mild,normal,TRUE,yes
overcast,mild,high,TRUE,yes
overcast,hot,normal,FALSE,yes
rainy,mild,high,TRUE,no
```

OUTPUT :

Classifier output

```

| humidity = normal: yes (2.0)
| outlook = overcast: yes (4.0)
| outlook = rainy
| windy = TRUE: no (2.0)
| windy = FALSE: yes (3.0)

Number of Leaves : 5
Size of the tree : 8

Time taken to build model: 0.01 seconds
```

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	7	50	%
Incorrectly Classified Instances	7	50	%
Kappa statistic	-0.0426		
Mean absolute error	0.4167		
Root mean squared error	0.5904		
Relative absolute error	87.5	%	
Root relative squared error	121.2987	%	
Total Number of Instances	14		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
Weighted Avg.	0.556	0.600	0.625	0.556	0.588	-0.043	0.633	0.758	yes
	0.400	0.444	0.333	0.400	0.364	-0.043	0.633	0.457	no

=== Confusion Matrix ===

```

a b  <-- classified as
5 4 | a = yes
3 2 | b = no
```

